

ANN Architecture

This document describes the trained ANN supplied for team review. Its saved weights are the same weights used for the reported test evaluation.

Selected configuration: weighted_regularised_32_16. Input layer: 16 encoded numerical features. Hidden layers: [32, 16] with ReLU activation. Output: one sigmoid neuron producing a churn score between zero and one. These scores have not been independently calibrated.

Dropout after each hidden layer: 0.2. L2 coefficient on hidden-layer kernels: 0.001.
Trainable parameters: 1089.

Adam learning rate 0.001; binary crossentropy; batch size 32; maximum 80 epochs; early stopping on validation loss with patience 10 and restored weights. Retained epoch: 41.
Balanced class weights used: True.

Training subset: 4313. Validation: 1079. Test: 1349. The StandardScaler was fitted only on the training subset. No refit was performed after selection.

Three configurations were compared across seeds [42, 43, 44]. Selection maximised mean validation F1 over thresholds 0.20 to 0.70 in steps of 0.05. Delivered seed: 42. Selected threshold: 0.45. Mean validation F1 for the selected pair: 0.5973; seed SD: 0.0140.

Pinned source commit: 6d1cb202e66568114f8d9a34a4cdeeb3dffb3. Input columns in order: SeniorCitizen, tenure, MonthlyCharges, EstimatedTotalSpend, NumActiveServices, AvgMonthlyChargePerService, gender_Male, Dependents_Yes, PhoneService_Yes, MultipleLines_Yes, InternetService_Fiber optic, Contract_One year, Contract_Two year, TenureGroup_Established, TenureGroup_Loyal, TenureGroup_Veteran

Inference requires encoded unscaled features. Use predict_churn.py with ann_final.keras, scaler_parameters.npz and metadata.json. Do not apply the scaler to already-standardised data.

The original preprocessing and duplicate-removal policy are inherited. A customer identifier is not available. Generalisation to new populations and causal effects have not been established.